

Organic Compounds Introduction

Understanding of organic compounds builds on material from previous modules

- Covalent bonding
 - Lewis structures
 - Electronegativity
 - Polar covalent bonding
 - VSEPR, molecular shapes, polar and non-polar molecules
 - Intermolecular forces
 - Dispersion forces
 - Dipole-dipole attraction
 - Hydrogen bonding
- The molecular structure of organic compounds determines their physical properties and reactivity
 - Understanding organic compounds stems from an understanding of their molecular structure

Definitions

- **ORGANIC COMPOUNDS**

- Compounds that are based on a framework of **carbon** atoms

- **ORGANIC CHEMISTRY**

- Study of the structure, properties and reactions of organic compounds

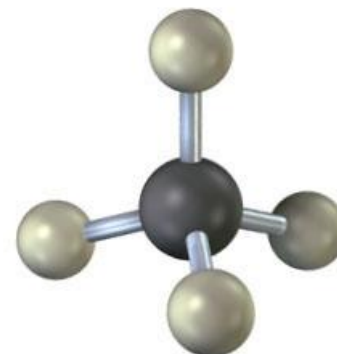
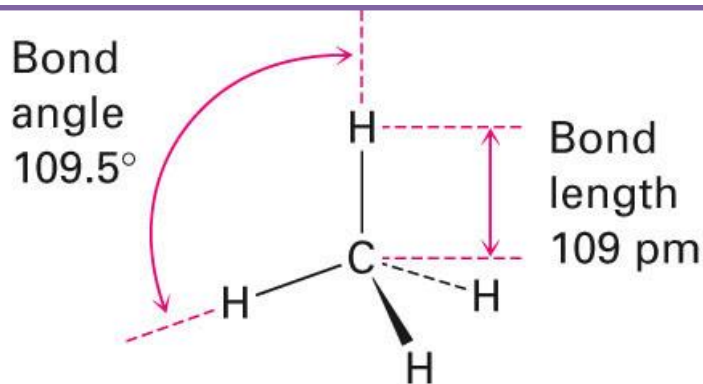
Organic Compounds

- This periodic table shows the elements most commonly observed in organic compounds
- We will focus on organic compounds that contain **carbon, hydrogen, oxygen, nitrogen and the halogens**
- It's useful to remember the number of valence electrons and thus the number of covalent bonds that these atoms tend to form

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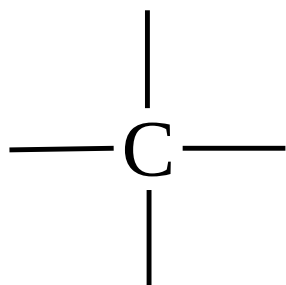
Carbon Atoms and Bonding

- Carbon is in Group 4 (14) of the periodic table
- It has 4 valence electrons
- To complete its octet, carbon tends to form 4 covalent bonds
- The simplest carbon compound CH_4 is shown below
- Note the tetrahedral arrangement of the electron pairs resulting in bond angles of 109.5°

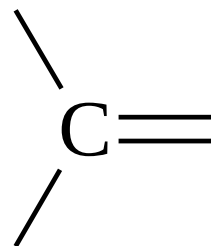


Carbon Bonding Combinations

The four bonds to carbon can be...



4 single bonds
Tetrahedral geometry



1 double bond and 2 single bonds
Trigonal planar geometry



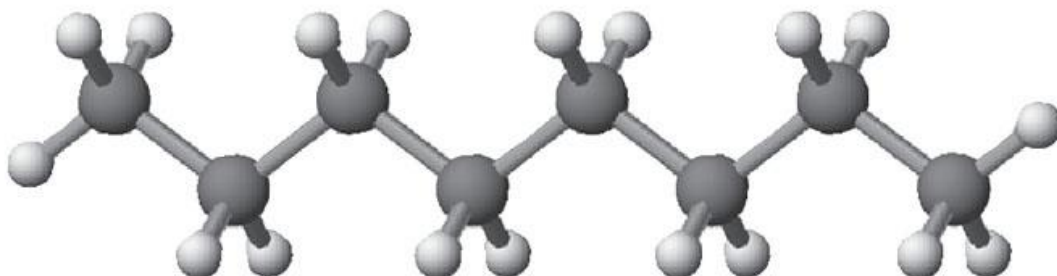
1 triple bond and 1 single bond
Linear geometry



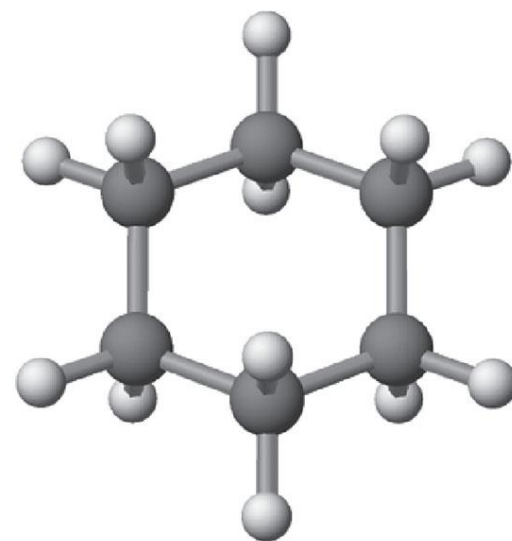
2 double bonds
Linear geometry

Carbon Bonding

- A distinctive property of carbon atoms is the ability to form bonds to other carbon atoms
- Chains and rings of carbon atoms are possible
- Carbon can also form bonds with other elements including hydrogen, oxygen, nitrogen and the halogens
- This results in the great diversity of organic compounds



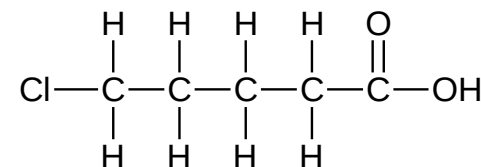
•Note that the tetrahedral arrangement of single bonds around carbon atoms also applies when carbon atoms form chains and rings



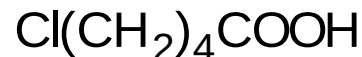
This lecture will describe how the structures shown here all represent the same compound

- Different organic compounds can have the same molecular formula
- They are different compounds because they have different structures – the atoms are connected in different arrangements
- To properly describe an organic compound, its structure must be shown in representations

Structural formula
(Lewis structure)

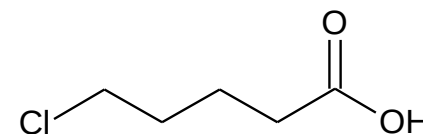


Condensed
formulas



Compound name: 5-chloropentanoic acid
Molecular formula: $\text{C}_5\text{H}_9\text{ClO}_2$

Line bond formula



Hydrocarbons Summary

Hydrocarbons

Alkanes, cycloalkanes and alkenes

Aromatic hydrocarbons

Hydrocarbons

- Hydrocarbons are compounds that contain **only carbon and hydrogen**
- In hydrocarbons the carbon atoms can be bonded in different ways
- Saturated hydrocarbon
 - Carbon atoms linked to others by single bonds
- Unsaturated hydrocarbon
 - One or more carbon atoms linked to others by a double or triple bond

Alkanes

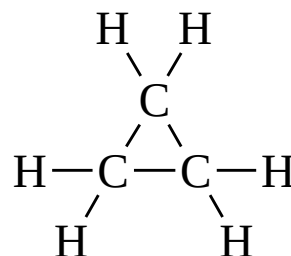
- Alkanes are **saturated hydrocarbons**
- Only C–C single bonds are present
- The other bonds to carbon are bonds to hydrogen atoms
- General formula C_nH_{2n+2}
- The names of alkanes end with 'ane'
- Alkanes can be considered to be the 'parents' of all organic compounds
- Develop knowledge of alkanes by doing the learning activities outlined in the Learning Guide
- You need to learn about:

- Isomerism of alkanes
- Explain the physical properties of alkanes

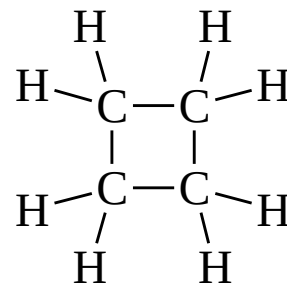
- Reactions of alkanes (combustion and halogenation)
- Recognise and name compounds that are alkanes

Cycloalkanes

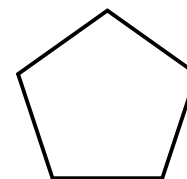
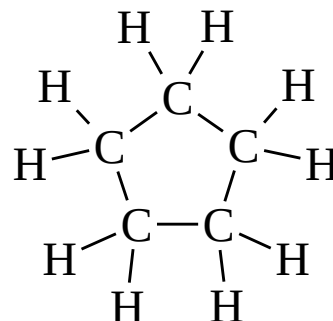
- Cyclic alkanes are also possible
- General formula C_nH_{2n}
- Cycloalkanes with $n = 3, 4$ and 5 are shown using structural formulas and line bond formulas
- Named using the same convention as straight chain alkanes, but with a prefix 'cyclo'



cyclopropane



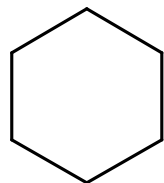
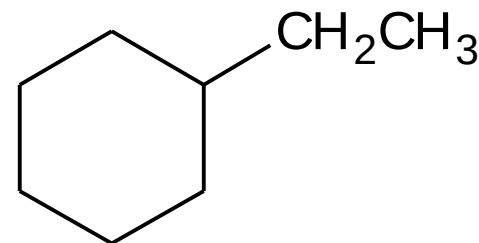
cyclobutane



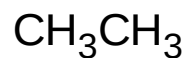
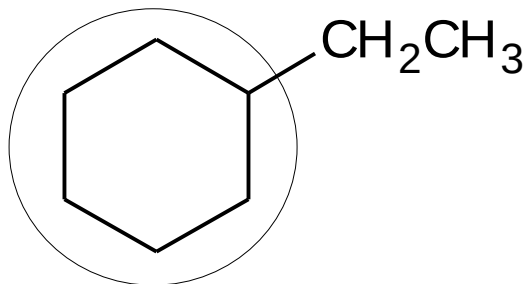
cyclopentane

Alkyl Groups

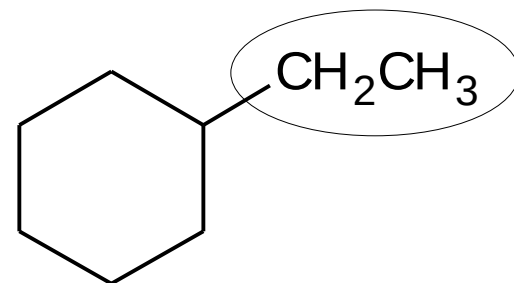
- Branching chains that occur in the structures of organic compounds
- Described using names that reflect the number of carbon atoms in the group
- Consider this example



cyclohexane



ethane



- This parent compound name is **cyclohexane**
- The circled alkyl group is called **cyclohexyl**

- This parent compound name is **ethane**
- The circled alkyl group is called **ethyl**

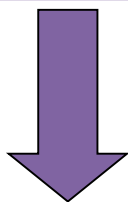
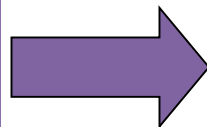
Alkenes

- Alkenes are **unsaturated hydrocarbons**
- They contain one or more carbon-carbon double bonds (C=C)
- General formula C_nH_{2n} (if there is one double bond)
- Names end with 'ene'
- Develop knowledge of alkenes by doing the learning activities outlined in the Learning Guide
- You need to be able to recognise alkenes and learn about the following:

- How to name compounds that are alkenes
- Structural isomerism of alkenes
- Geometric isomerism of alkenes
- Addition reactions of alkenes

Aromatic Hydrocarbons

Hydrocarbons



AROMATIC HYDROCARBONS

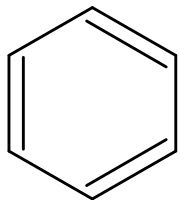
- Unsaturated
- Special properties
- Contain a benzene ring

ALIPHATIC HYDROCARBONS

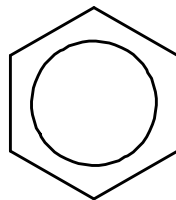
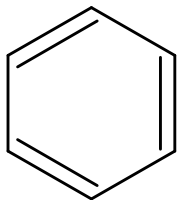
- Alkanes – saturated
- Alkenes – unsaturated
- Aliphatic comes from a Greek word 'aleiphas' meaning fat
- These compounds are oily by nature and this is related to their molecular structure

Benzene

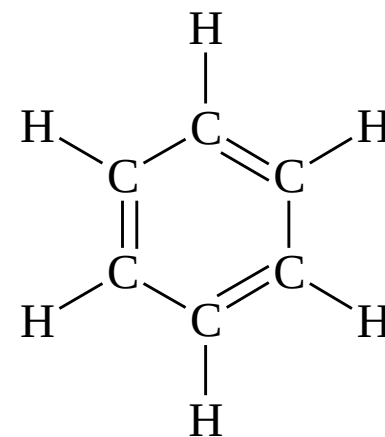
- Benzene is the parent of all aromatic compounds
- Molecular formula C_6H_6
- Different representations are shown
- Note that 2 correct structures can be drawn (resonance structures) and this is shown for the line bond formula



Line bond formula



Alternative line bond representation

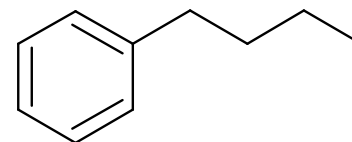
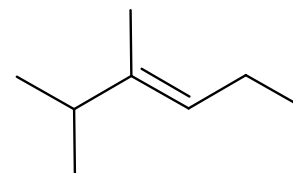
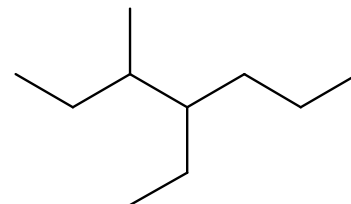


Structural Formula

- Any compound that contains a benzene ring is aromatic
- You must be able to recognise aromatic compounds
- Name simple aromatic compounds

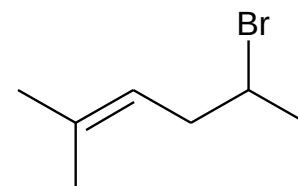
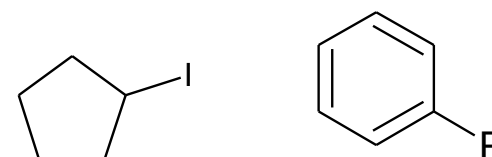
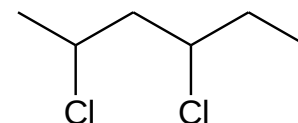
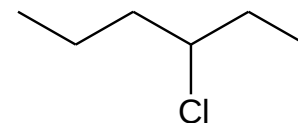
This lecture will demonstrate how to apply the steps and name the following hydrocarbons

Step	Description of the step
0	Examine the structural representation to identify the type or classification of the organic compound
1	Identify the parent chain, which is the longest continuous chain of carbon atoms
2	Number the carbon atoms in the parent chain
3	Identify the substituents and the number of the parent chain carbon atoms that they branch from
4	Write the whole name of the compound as one word



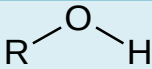
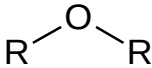
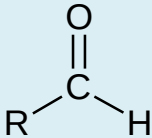
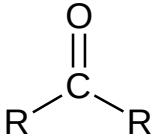
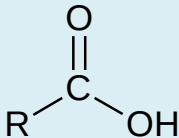
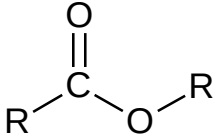
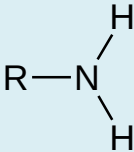
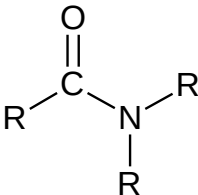
This lecture will demonstrate how to apply the steps and name these compounds that contain halogen atoms

Step	Description of the step
0	Examine the structural representation to identify the type or classification of the organic compound
1	Identify the parent chain, which is the longest continuous chain of carbon atoms
2	Number the carbon atoms in the parent chain
3	Identify the substituents and the number of the parent chain carbon atoms that they branch from
4	Write the whole name of the compound as one word



Functional Groups

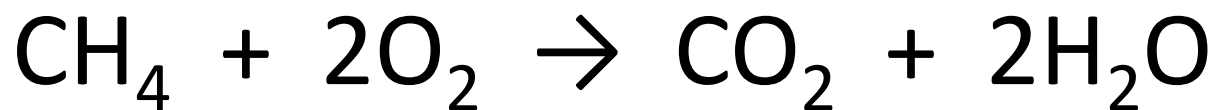
- Organic compounds can be classified according to the presence of functional groups
 - An atom or group of atoms attached to a hydrocarbon framework
 - Predictions about organic compounds can be made based on the presence of particular functional groups
- **FUNCTIONAL GROUPS** studied in this module are detailed on the following slide
 - The presence of a particular functional group affects physical properties and chemical reactivity of organic compounds
 - Understanding related to this will develop through the module
- **R GROUPS**
 - You will see the letter 'R' used in structural formulas of organic compounds
 - R can be any hydrocarbon framework (straight or branched chain of carbon atoms, ring of carbon atoms, benzene ring)
 - If more than one R group is shown, they can be the same or different

Name	General structure	Condensed formula	Suffix in naming
Alcohol		ROH	-ol
Ether		ROR	Naming not covered
Aldehyde		RCHO	-al
Ketone		RCOR	-one
Carboxylic acid		RCOOH	-oic acid
Ester		RCOOR	-ate
Primary amine		RNH ₂	-amine
Amide		CONR ₂	Naming not covered

Reactions of Organic Compounds Summary

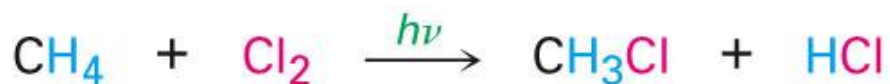
Combustion of Alkanes

- Alkanes are fairly inert and unreactive substances
- They can react with oxygen and halogens
- The reaction of alkanes with oxygen occurs during **COMBUSTION**
- This reaction occurs with the release of heat energy and alkanes are thus used as fuels
- The complete combustion of methane (natural gas) is shown in the equation



Halogenation of Alkanes

- In the presence of sunlight or an ultraviolet (UV) light, alkanes can react with halogens (eg chlorine and bromine)
- The reaction between methane and chlorine is shown in the equations below
- The $h\nu$ above the reaction arrow denotes that light is required for the reaction



- In each step, a hydrogen atom is substituted by a chlorine atom
- Example of a substitution reaction

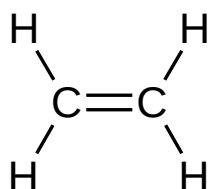
- The analogous reaction occurs with Br_2



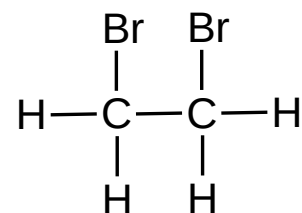
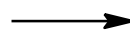
Addition Reactions of Alkenes

- For alkenes, the carbon-carbon double bond is a site for chemical reactivity
- One of the major reactions of alkenes is the addition of a small molecule to the double bond
- These belong to a broad class of organic reactions called **ADDITION REACTIONS**

- Addition of Br_2 to an alkene double bond is shown here
- However, other small molecules can also add to an alkene double bond (eg H_2O , HBr , H_2 , X_2)



+



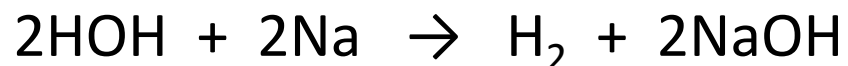
Ethene
Unsaturated

1,2-dibromoethane
Saturated

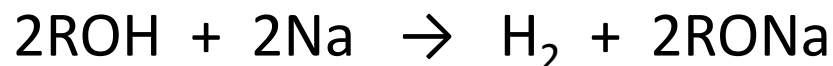
Reactions of Alcohols

Reaction with Group I Metals

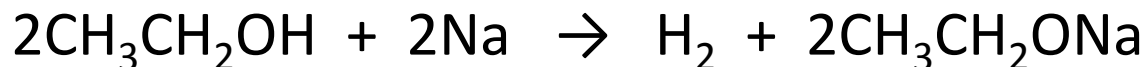
- Group I metals react with water to produce hydrogen gas and alkali metal hydroxides
- This reaction is shown for sodium



- Alcohols react with Group I metals in the same manner, see the general equation to the right

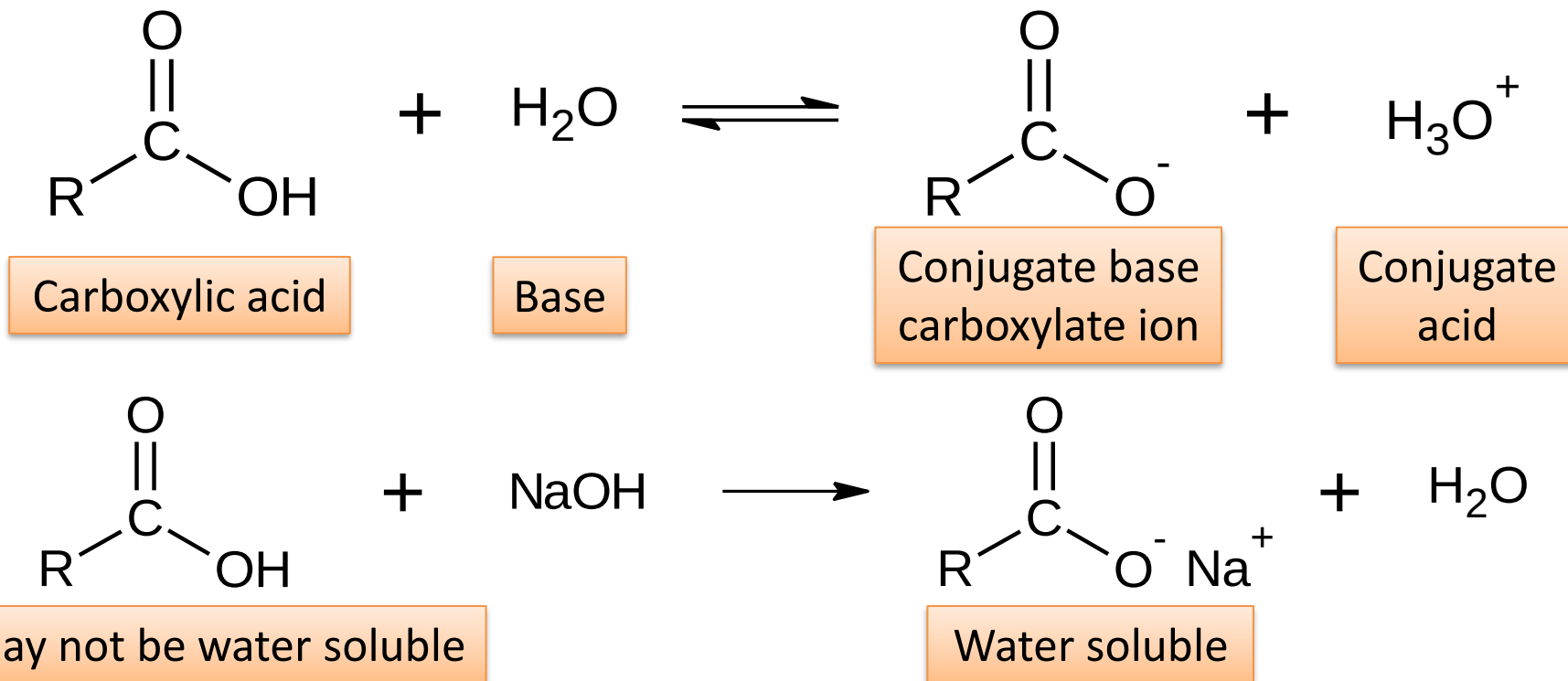


- The equation for the reaction of ethanol with sodium is shown below
- $\text{CH}_3\text{CH}_2\text{ONa}$ is called sodium ethoxide, which is a salt composed of $\text{CH}_3\text{CH}_2\text{O}^-$ and Na^+ ions



Acidity of Carboxylic Acids

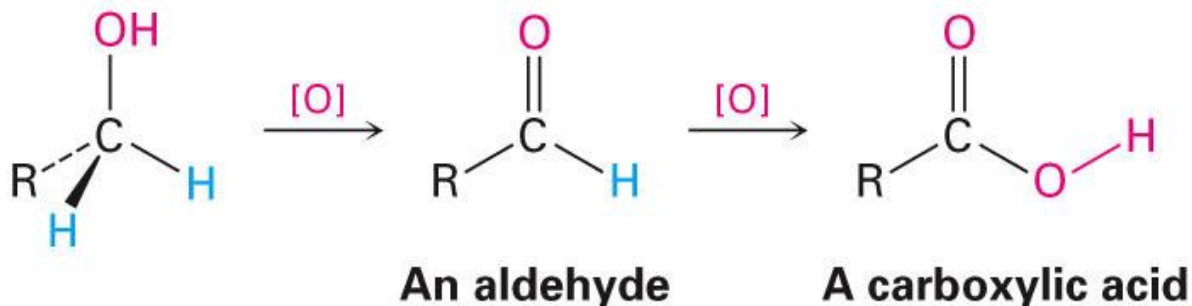
- The acidity of carboxylic acids is shown in the equations below
- Weak acids do not dissociate completely when added to water
- Incomplete dissociation shown in equations using reversible reaction arrows
- React with strong bases in the same manner as strong acids



Oxidation of Alcohols

- The outcome of the oxidation depends on the structure of the starting alcohol
- Primary alcohols are oxidised initially to aldehydes and then oxidation continues
- The final product is a carboxylic acid

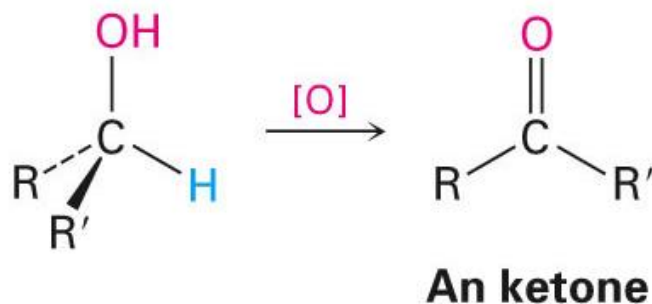
Primary alcohol



•Note – $[\text{O}]$ is a general representation to denote oxidation, in each of the alcohol oxidations shown the oxidising agent could be acidified permanganate (KMnO_4 , H^+)

Oxidation of Alcohols

Secondary alcohol



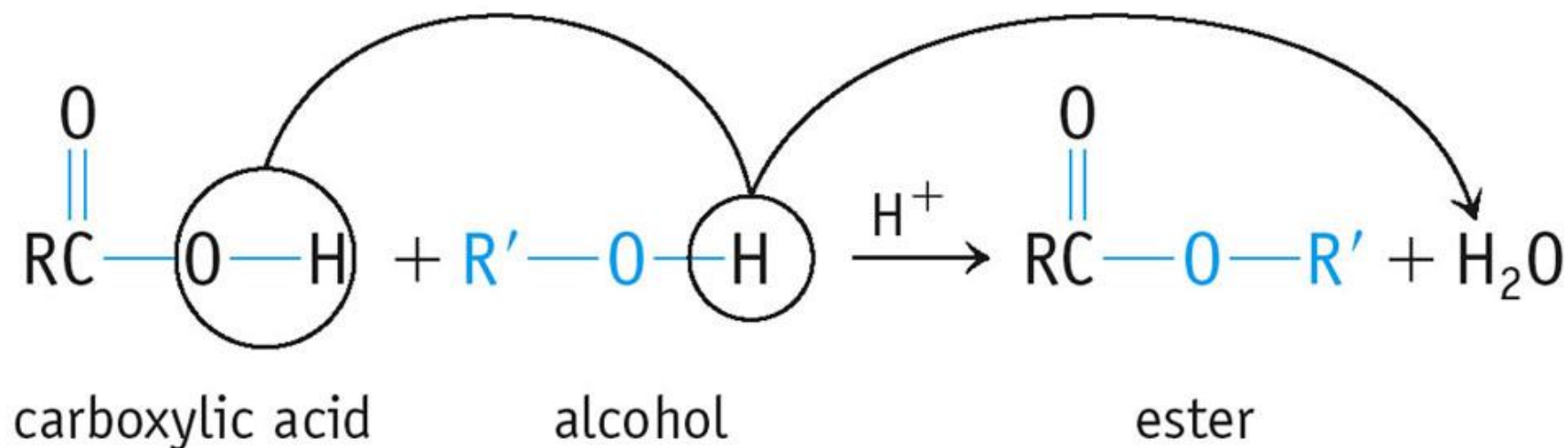
- Secondary alcohols are oxidised to ketones
- Tertiary alcohols are resistant to oxidation

Tertiary alcohol



Formation of Esters

- The reaction between a carboxylic acid and an alcohol gives an ester as the product
- An acid catalyst is required for the reaction
- Example of a **CONDENSATION REACTION** – a reaction that involves combining of two molecules and a small stable molecule is expelled
- The small molecule expelled in this particular example is a water molecule

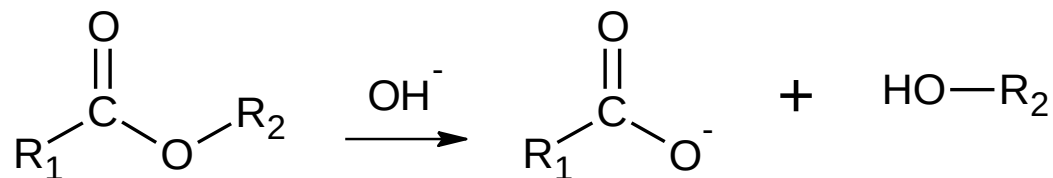


Hydrolysis of Esters

- **Hydrolysis** is a chemical reaction of a compound with water

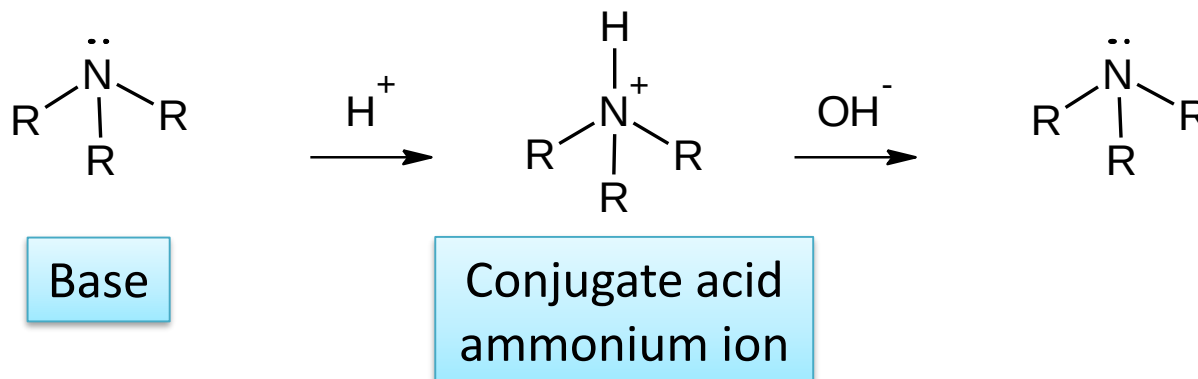
- Esters can undergo hydrolysis but the reaction with pure water is very slow
- Faster rates in the presence of acid or base (we focus on the outcome in basic conditions)

- In the presence of **base**
- Products: alcohol and carboxylate ion
- This is the reaction that occurs in the production of soap from fats

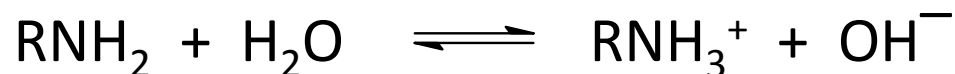


Basicity of Amines

- The non-bonding pair (lone pair) of electrons present in amine molecules enables them to accept a hydrogen ion (H^+)
- The lone pair electrons are used to form a covalent bond with a hydrogen ion (proton)

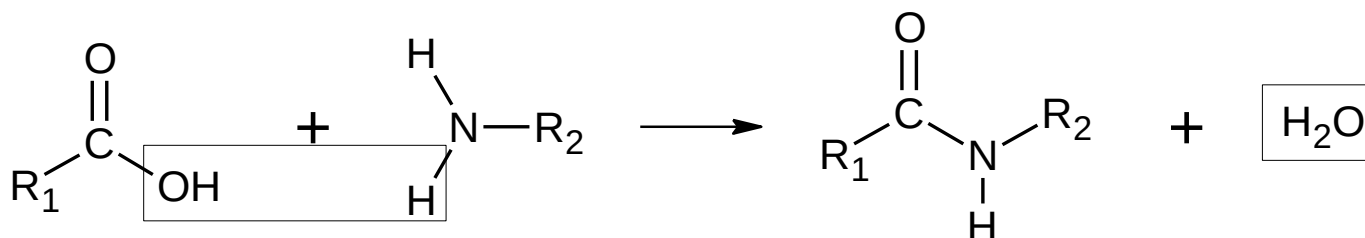
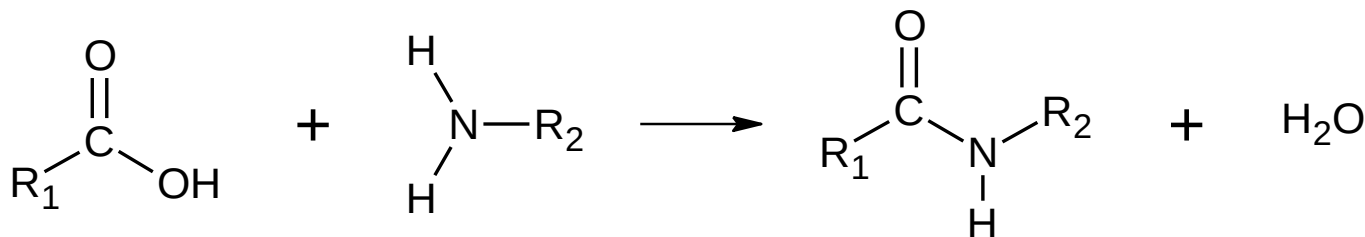
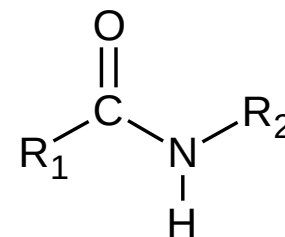


- Aqueous solutions of amines are basic because of the reaction with water shown on the right
- Amines are weak bases



Formation of Amides

- The general structure of the **amide** functional group is shown to the right
- Amides can form in a condensation reaction between a carboxylic acid and an amine
- Amides occur in proteins and some polymers



Physical Properties of Organic Compounds – Summary

Physical properties are related to intermolecular forces (IMF)

- The physical properties of substances depend on the type of **INTERMOLECULAR FORCES** that occur between the individual molecules that make up the substance
- The intermolecular forces that are possible depend on molecular structure
- To fully appreciate this you need to have a good understanding of
 - dispersion forces
 - dipole-dipole attraction
 - hydrogen bonding

Intermolecular forces for organic compounds

Dispersion forces

- Possible for all organic compounds
- Only possible IMF for non-polar compounds such as hydrocarbons
- Weakest IMF in relative terms, but strength increases with molecular size

Dipole-dipole attraction

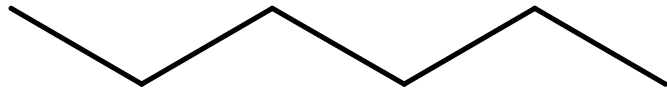
- Possible for polar organic compounds such as ethers, aldehydes, ketones, esters

Hydrogen bonding

- Possible for organic compounds that have highly polar O–H and N–H bonds
- For example: alcohols, carboxylic acids, amines, amides

Physical properties are related to intermolecular forces (IMF)

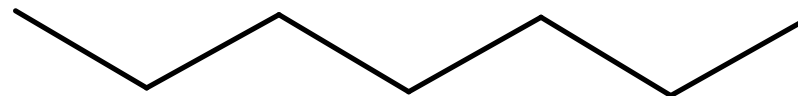
- You must be able to consider the molecular structure of an organic compound and determine what type of IMF can occur



hexane C_6H_{14}
BP = 69 °C

- Hexane and heptane are both non-polar, straight chain alkanes
- Dispersion forces only IMF possible for both compounds

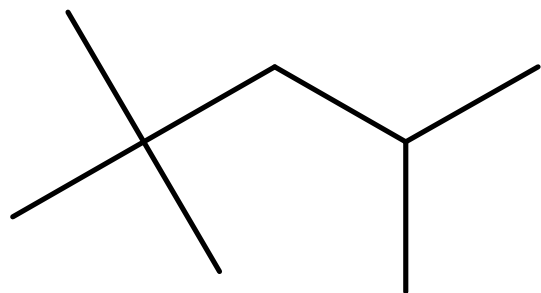
- Explain physical properties of organic compounds
- Explain differences between physical properties of organic compounds



heptane C_7H_{16}
BP = 98 °C

- Explain the difference in boiling point between the two compounds
- Dispersion forces are stronger for the compound with higher molar mass

Physical properties are related to intermolecular forces (IMF)



2,2,4-trimethylpentane
(iso-octane) C_8H_{18}
BP = 99 °C

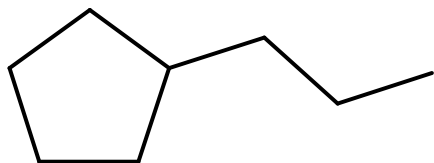


octane C_8H_{18}
BP = 125 °C

- These compounds are structural isomers (same molar mass)
- Both non-polar alkanes
- Dispersion forces only IMF possible for both compounds

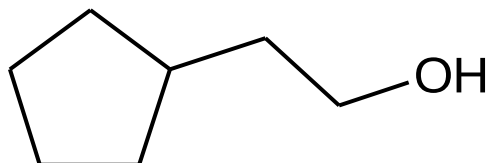
- Explain the difference in boiling point between the two compounds
- Dispersion forces are weaker for the branched isomer
- Due to decreased surface area

Arrange the following compounds that have similar structures and molar mass in order of expected boiling point from lowest to highest



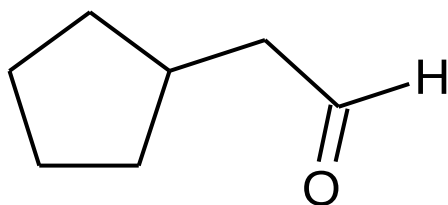
1) 112 g mol^{-1}

Alkane
Dispersion forces only
Weakest IMF



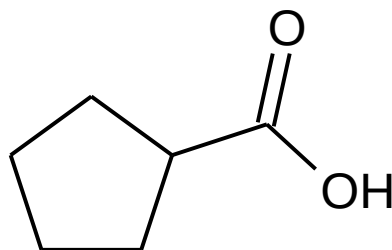
2) 114 g mol^{-1}

Alcohol
Hydrogen bonding
Strongest IMF



3) 112 g mol^{-1}

Aldehyde
Dipole-dipole attraction
Stronger than dispersion
Not as strong as H-bonding



4) 114 g mol^{-1}

Carboxylic acid
Hydrogen bonding
Stronger than alcohol

- The small difference in molar mass is not the reason for any difference in boiling point
- The difference is due to strength of the intermolecular forces
- Lowest boiling point expected for the compound with the weakest IMF
- Highest boiling point expected for the compound with the strongest IMF

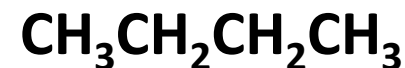
Expected order of boiling point $1 < 3 < 2 < 4$

Water solubility of organic compounds is another physical property that can be predicted

- Water is a polar substance that can dissolve other polar substances
- Very good solvent for compounds that can participate in H-bonding

- Lowest water solubility expected for the least polar compound
- Highest water solubility expected for the most polar compound

Arrange the following compounds in order of expected water solubility from lowest to highest

60 g mol⁻¹59 g mol⁻¹60 g mol⁻¹58 g mol⁻¹

Ester – polar due to the presence of polar bonds and molecular shape

Amine – polar and can participate in H-bonding interaction with water molecules

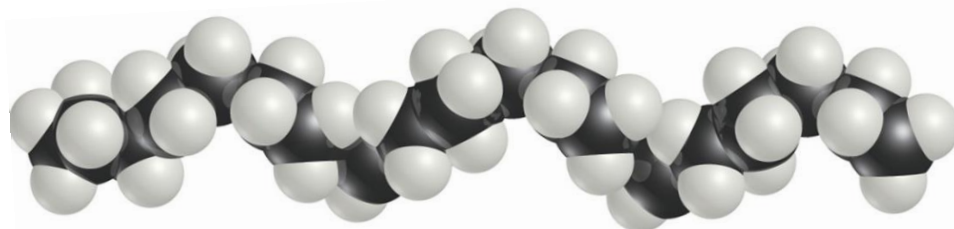
Carboxylic acid – polar and can participate in H-bonding with water, more polar than amine

Alkane – non-polar

Expected water solubility CH₃CH₂CH₂CH₃ < HCOOCH₃ < CH₃CH₂CH₂NH₂ < CH₃COOH

Polymers

- Polymers are large molecules made up of repeating units
- Composed of small molecules called **monomers**
- Polymerisation is the name of the process in which monomers react to form polymers



Natural polymers

Rubber, proteins, DNA, wool, cellulose, starch

Synthetic polymers

Polystyrene, PVC, Teflon, nylon,

Homopolymer – one type of monomer

Copolymer – two different monomers

ADDITION POLYMERISATION

- Alkene monomers react with each other to form a polymer
- Polymerisation involves repeating addition reactions
- Produces homopolymers called addition polymers

CONDENSATION POLYMERISATION

- Monomers have two reactive functional groups
- Polymerisation involves condensation reactions between monomers
- Produces copolymers called condensation polymers

Condensation polymers

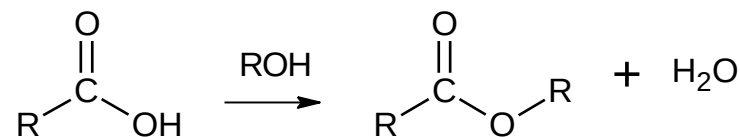
Recall from the chemistry of the functional groups that condensation reactions can be used to produce esters and amides

POLYESTER MONOMERS

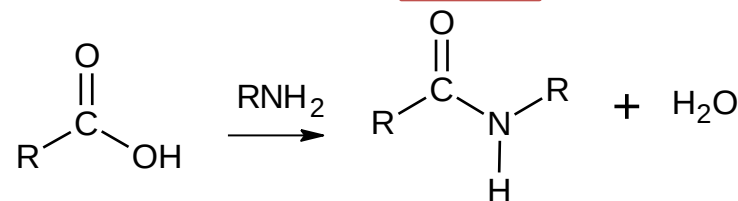
- dicarboxylic acid and diol

POLYAMIDE MONOMERS

- dicarboxylic acid and diamine

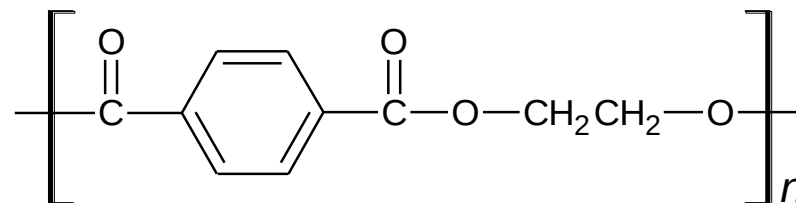
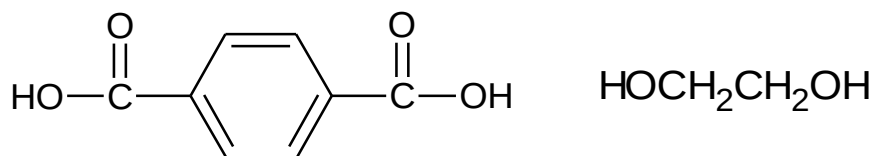


ester

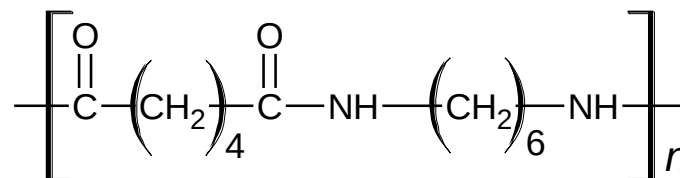
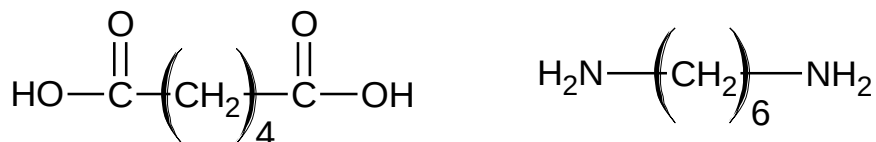


amide

A polyester and its monomers



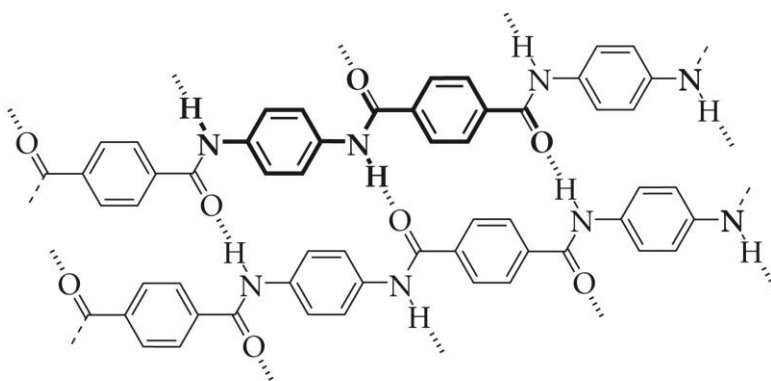
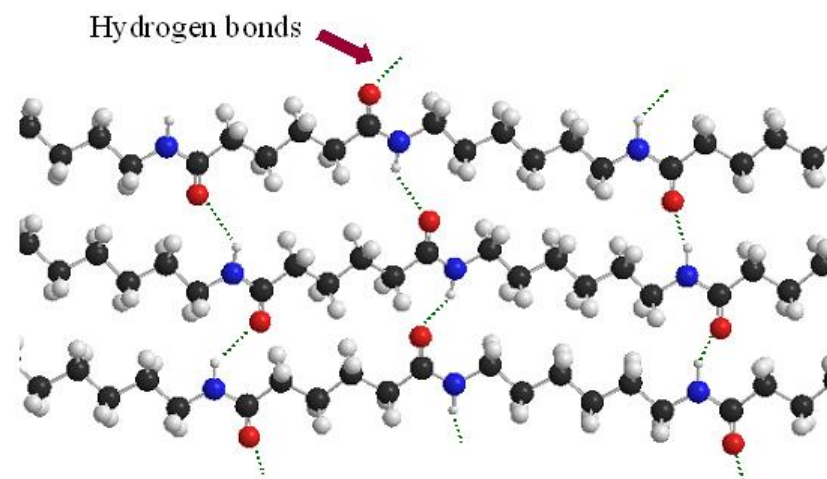
A polyamide and its monomers



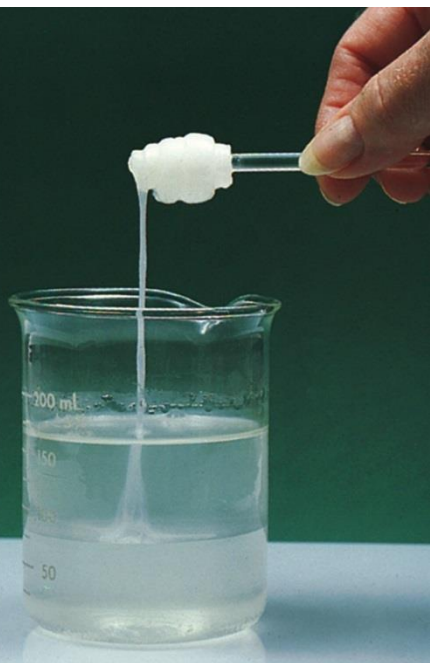
Draw repeating units of condensation polymers, given the structure of the monomers
Examine the structure of a condensation polymer, identify and draw the monomer structures

Condensation polymers

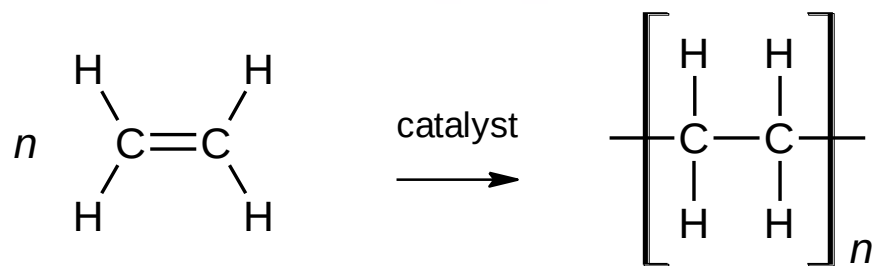
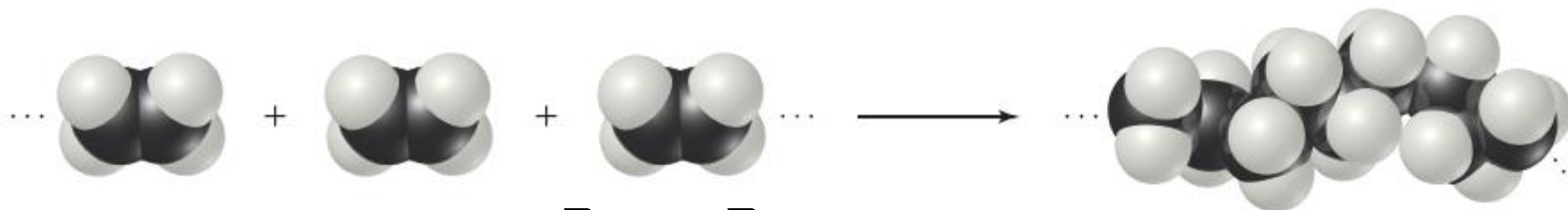
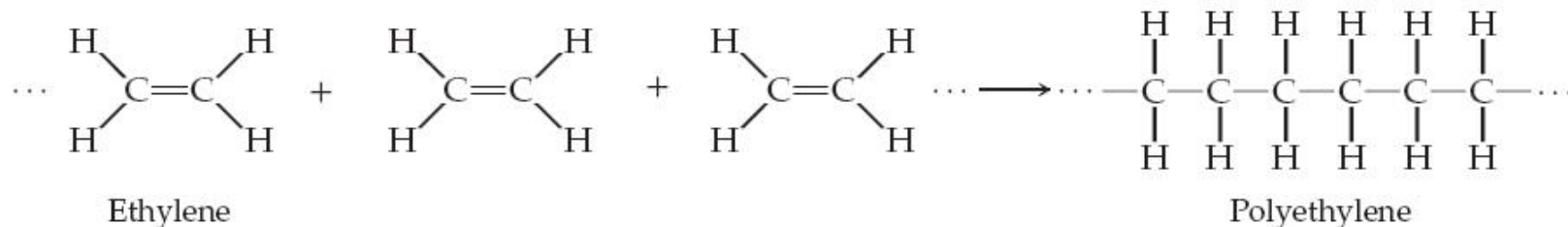
- The polyamide shown on the previous slide is actually nylon, which forms at the interface of the two liquid reactants (image below)
- The properties of nylon are due to the intermolecular forces between the polymer chains that make up the material (shown on the right)



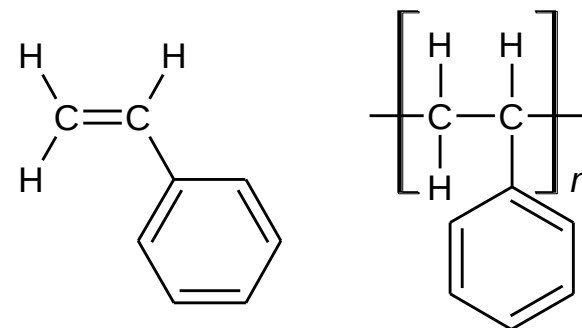
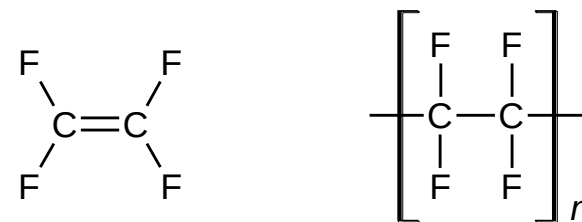
- Kevlar is a polyamide
- Structure shown to the left
- Very strong material
- Properties related to molecular structure and intermolecular forces between the polymer chains



Addition polymers



Two addition polymers and their monomers

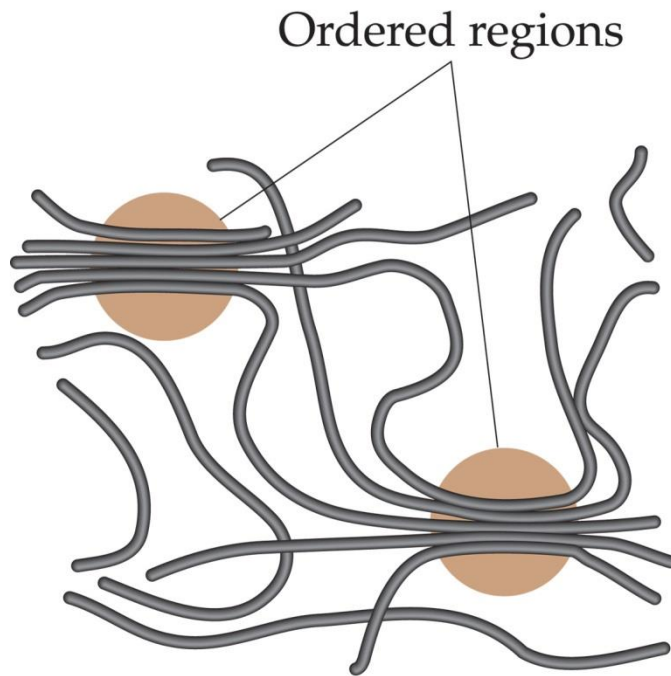


- Since polymerisation is a chemical reaction it can be represented using equations
- Note the use of brackets to show the repeating units of a polymer
- n = hundreds to thousands
- Examine representations of addition polymers, identify the monomer and draw its structure
- Draw a repeating unit for an addition polymer

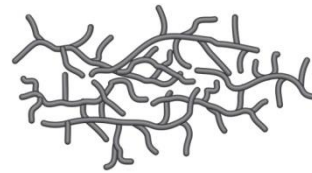
Addition polymers

- Materials known as polymers are made up of many polymer chains of varying lengths formed in the polymerisation reaction
- The chains tangle and may interact in various locations by dispersion forces

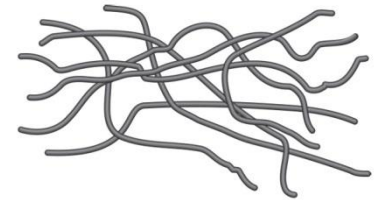
- Reaction conditions for polymerisation can be controlled to produce mainly linear or highly branching polymer chains
- This gives rise to materials with different properties
- For example, LDPE and HDPE



LDPE

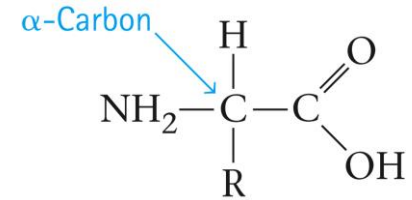


HDPE

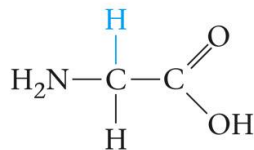


Amino acids

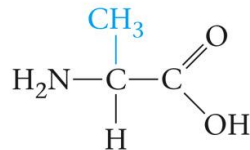
- **PROTEINS** are polymers of amino acids
- General structure of α -amino acids is shown to the right
- There are 20 amino acids that occur in most proteins
- They all have different R groups
- Note that each amino acid has a unique name and three letter code – you do not have to know these



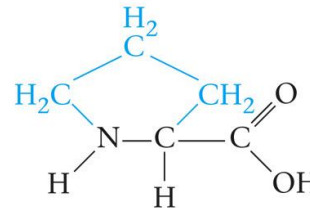
Nonpolar R Groups (hydrophobic)



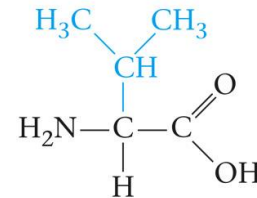
Glycine (gly)



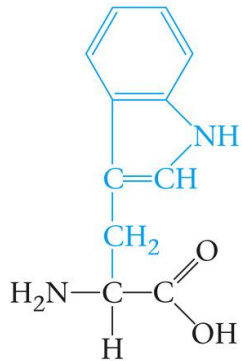
Alanine (ala)



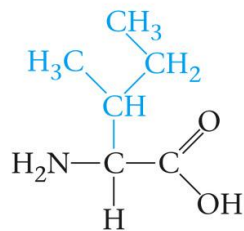
Proline (pro)



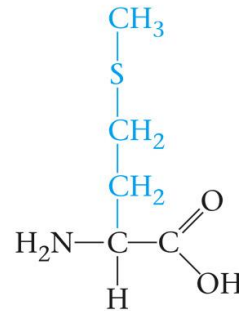
Valine (val)



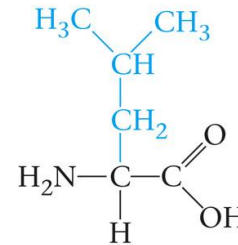
Tryptophan (trp)



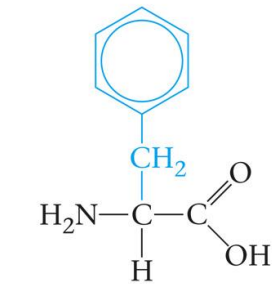
Isoleucine (ile)



Methionine (met)



Leucine (leu)

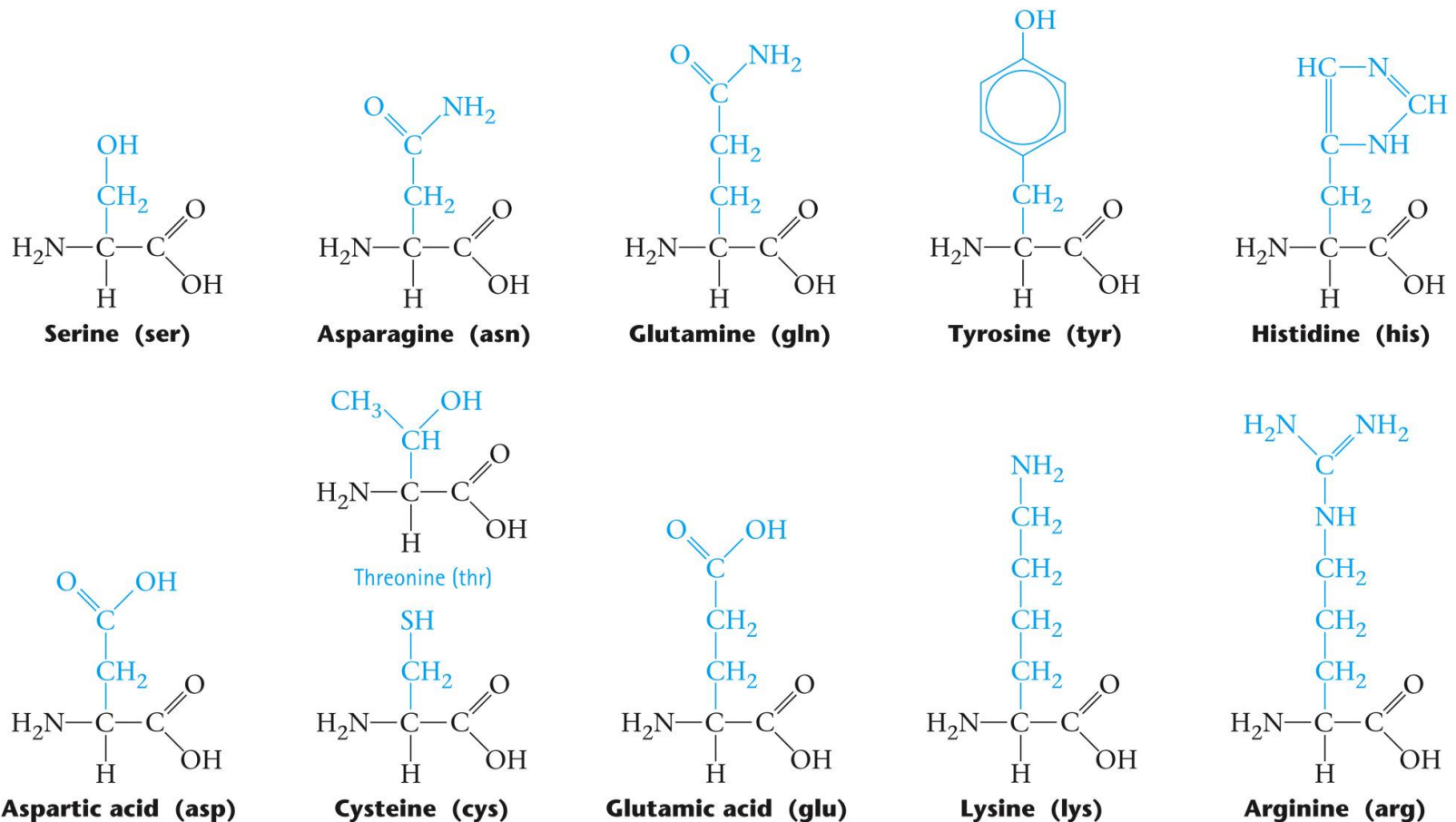


Phenylalanine (phe)

Amino acids

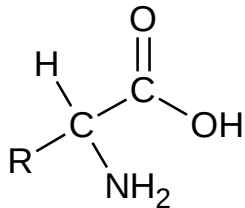
- The amino acid R groups play a role in the structure of proteins, and thus their properties and functions

Polar R Groups (hydrophilic)

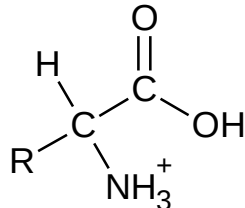


Amino acids are amphoteric and can exist in different forms

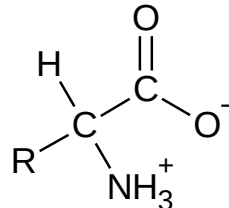
- A substance that can act as an acid or base is **amphoteric**
- Amino acids exist in different forms in aqueous solution depending on the pH



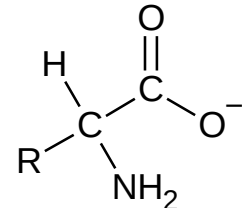
General
representation



Form in acidic
solution (pH < 7)



Form in neutral
solution (pH = 7)



Form in basic
solution (pH > 7)

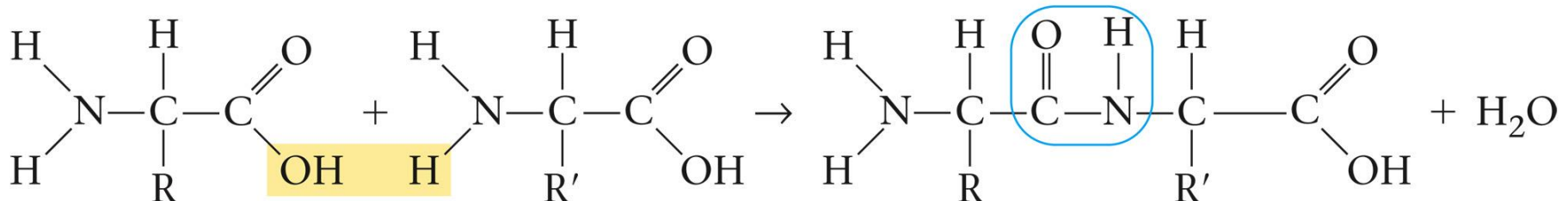
Zwitterion

- A molecule that is neutral overall, but one part is positively charged and another part is negatively charged
- Amino acids are zwitterions in solution at pH 7 and in the solid state
- Amino acids in the solid state have physical properties similar to ionic compounds

Proteins – peptide bonds

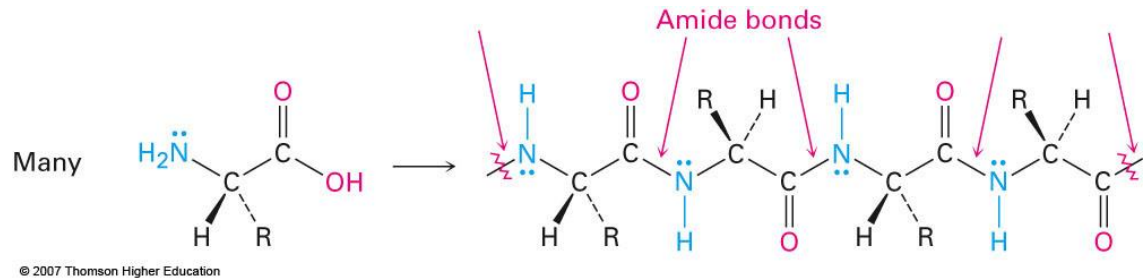
- The reactivity of carboxylic acid and amine functional groups means that amino acids can react with each other
- A condensation reaction between the amine and carboxylic acid functional groups of different amino acids can occur
- The functional group that forms in this reaction is an amide

- Peptide – any group of two or more amino acids linked together
- Dipeptide – 2 amino acids
- Oligopeptide – 3 to 10 amino acids
- Polypeptide – more than 10 amino acids (can be hundreds)
- The amide bonds in peptides are often referred to as peptide bonds or peptide linkages



Protein primary structure

- Proteins are essentially polyamides
- Protein structure can be described on four levels
- Primary, secondary, tertiary and quaternary



- PRIMARY STRUCTURE**
- The sequence of amino acids
- Maintained by the peptide bonds
- Determines the subsequent levels of structure
- Two examples of protein primary structure are shown to the right
 - oxytocin
 - vasopressin
- Small differences in sequence, but completely different functions

cys-tyr-**ile**-gin-asn-cys-pro-**leu**-gly

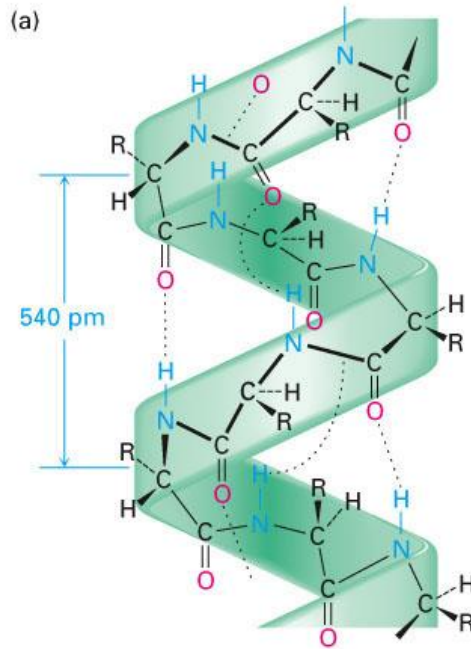
a

cys-tyr-**phe**-gin-asn-cys-pro-**arg**-gly

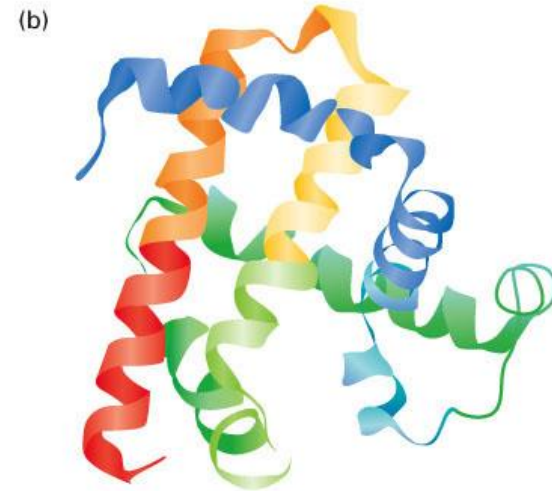
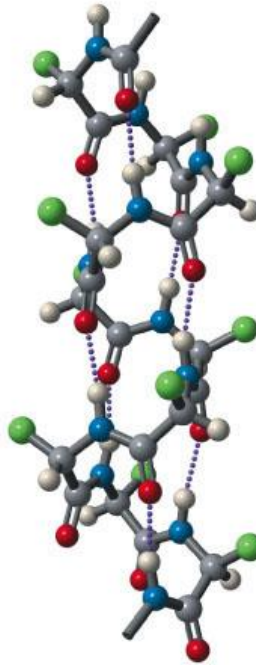
b

Protein secondary structure

α -helix



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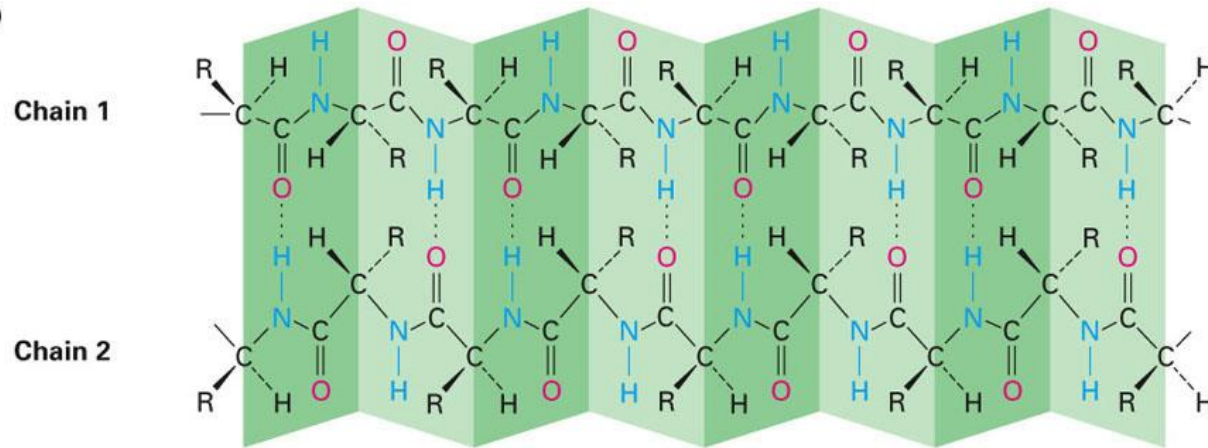
- The secondary structure of a protein are repeating patterns that occur along the chain
- The α -helix is a common example
- Coiling of the polymer chain

- Forms through attractions between the hydrogen atoms of the N–H groups and oxygen atoms in the C=O groups of the amide bonds in the chain
- The R groups point to the outside of the coil
- Figure (b) shows a protein that has extensive α -helices

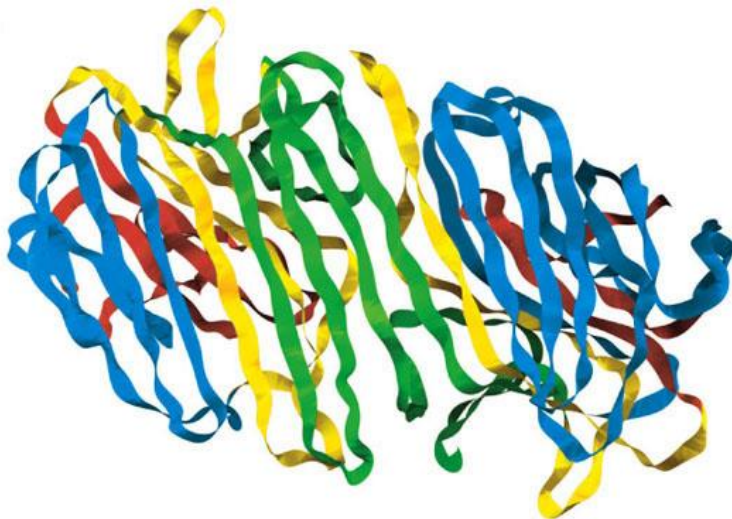
Protein secondary structure

β -pleated sheet

(a)

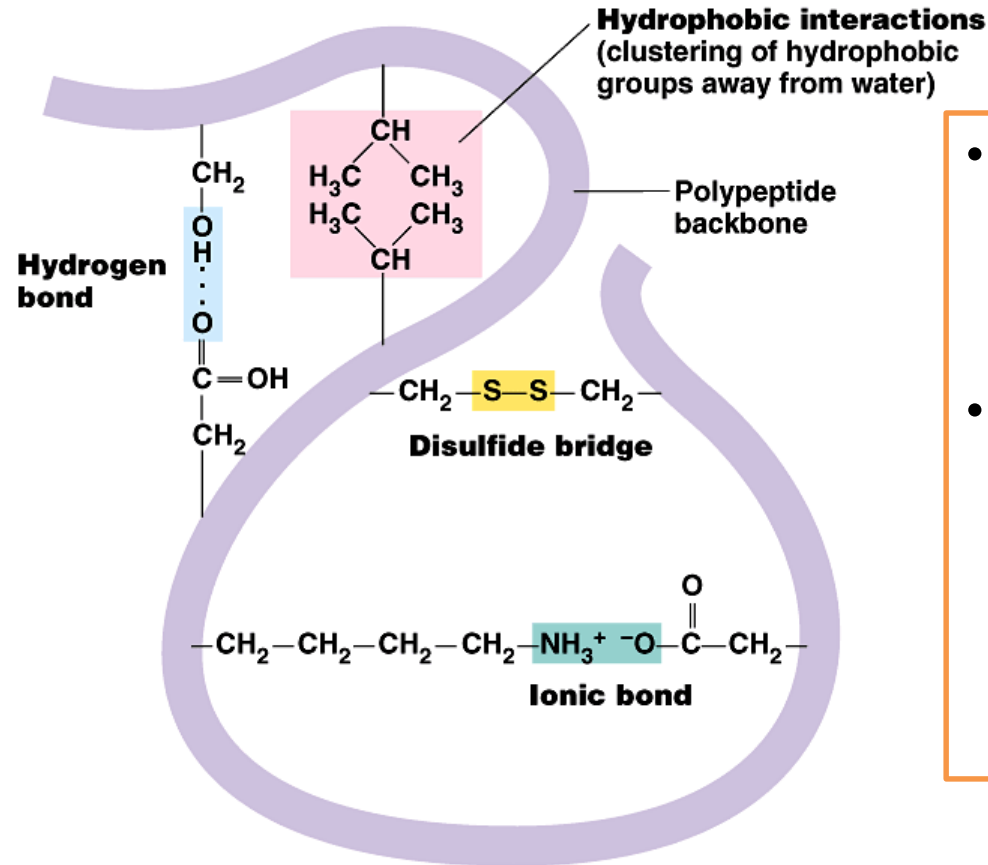


(b)



- The β -pleated sheet is another common example of protein secondary structure
- Adjacent polypeptide chains form a zigzag sheet through intermolecular interactions between the chains as shown in the figure (a)
- Figure (b) is an example of a protein that displays this structure extensively

Protein tertiary structure

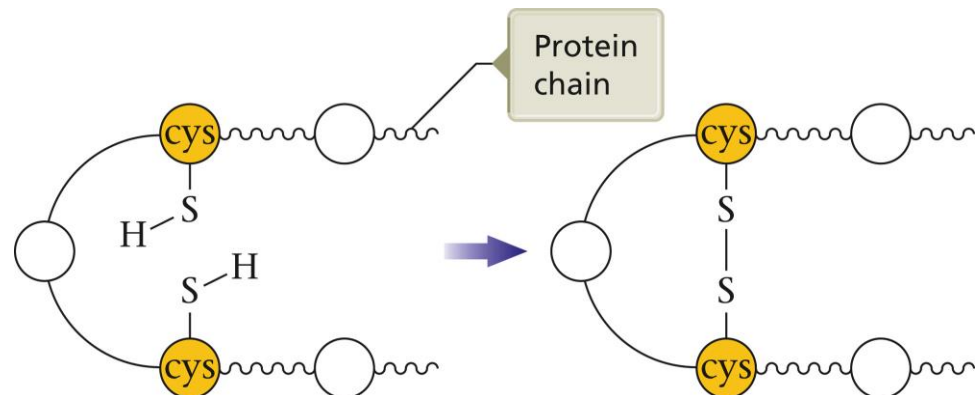


- Folding of the protein chain that occurs due to interactions between R groups far apart in the sequence of amino acids
- Occur through intermolecular or attractive forces
 - Hydrogen bonds
 - Hydrophobic interactions (dispersion forces)
 - Salt bridges (ionic bonds)

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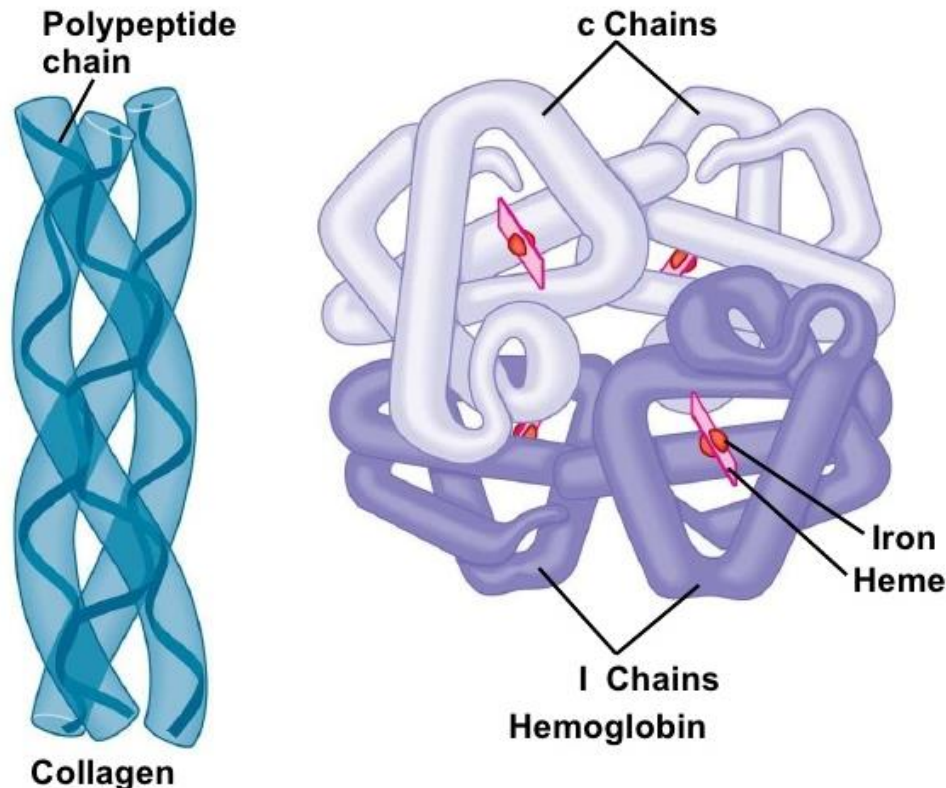
Disulfide linkage

- The amino acid cysteine (cys) has an $-SH$ group in the side chain
- Adjacent SH groups can react to form an $S-S$ covalent bond



Protein quaternary structure

- Many proteins are made up of subunits, which are individual chains that aggregate to form the functioning protein
- Quaternary structure is how the subunits are arranged
- The same types of interactions that maintain tertiary structure are responsible for quaternary structure



Levels of protein structure summary

Primary – sequence of amino acids

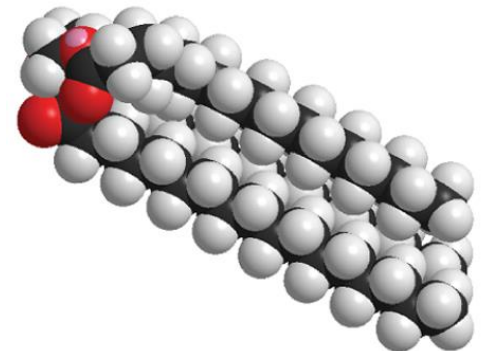
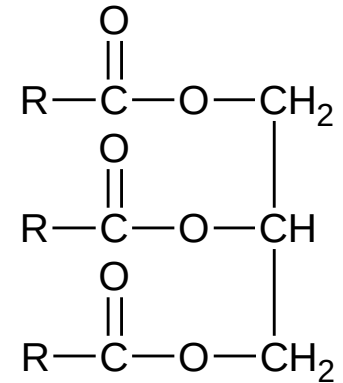
Secondary – repeating patterns along a polypeptide chain

Tertiary – folds in the polypeptide chain due to interactions between distant R groups

Quaternary – arrangement of subunits in proteins

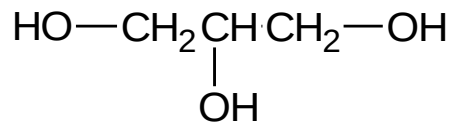
Fats

- The difference between fats and oils is merely in their physical state
- Fats are solids and oils are liquid at room temperature
- In terms of molecular structure they are esters, specifically they are triesters
- Also known as triglycerides
- Representations of fat molecules are shown
- The R groups in the general structure can be saturated or unsaturated hydrocarbon chains with 10 – 20 carbon atoms, they can be the same or different

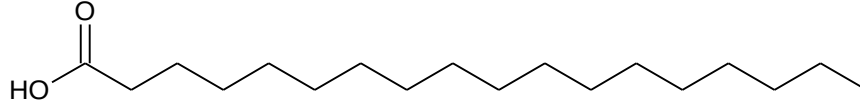


Fats

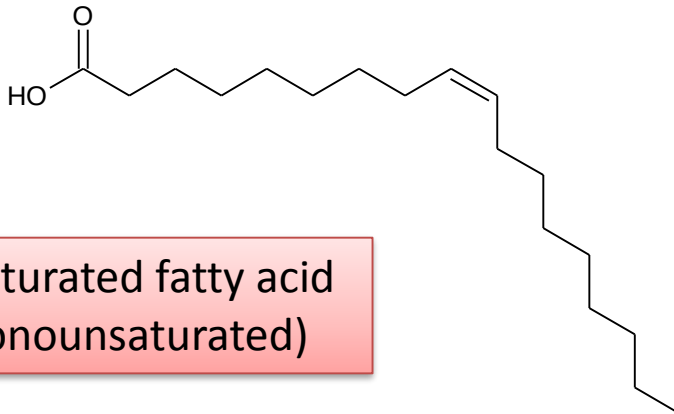
- Like all esters, fats are considered to be derived from an alcohol and a carboxylic acid
- The alcohol is a triol called **glycerol**
- The carboxylic acids have long hydrocarbon chains and are known as **fatty acids**



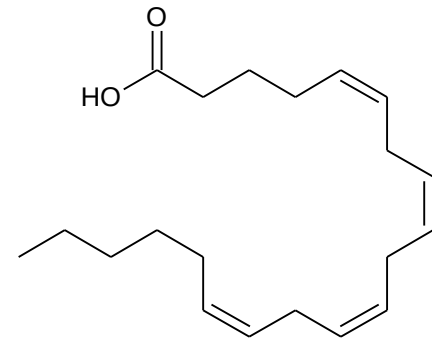
Glycerol – the parent alcohol of fat triesters



Saturated fatty acid



Unsaturated fatty acid
(monounsaturated)

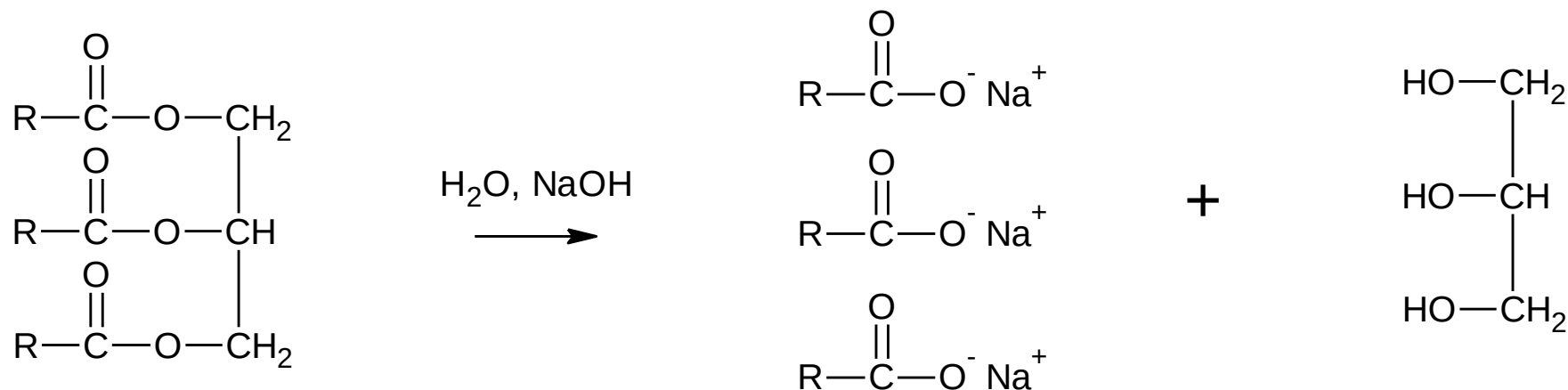


Unsaturated fatty acid
(polyunsaturated)

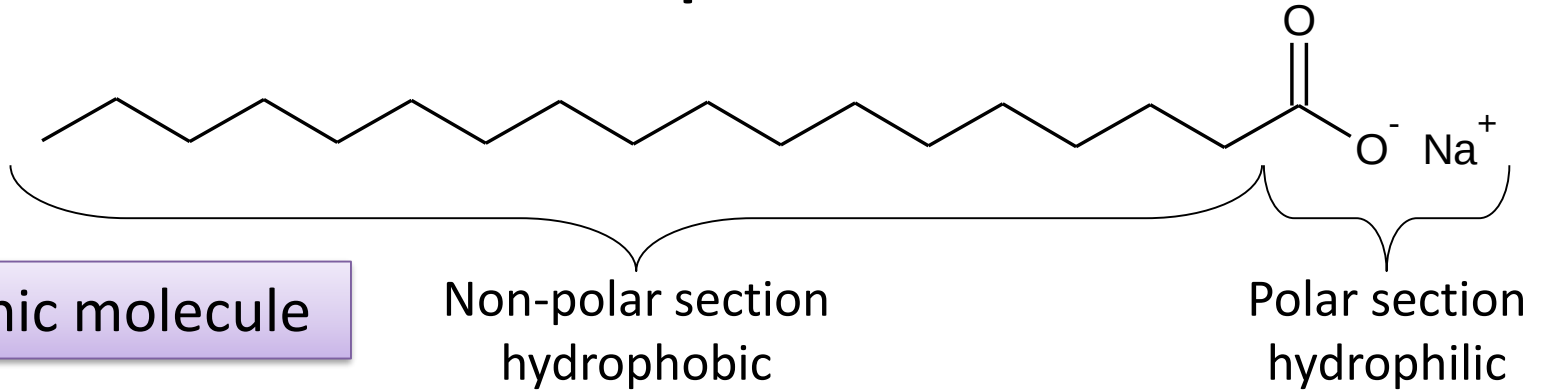
Soaps

- Recall that esters can undergo hydrolysis in the presence of base to form an alcohol and carboxylate ion
- This reaction is also called saponification
- The term saponification is generally used to describe the production of soap from fats

- The hydrolysis of fat triester molecules in the presence of base produces glycerol and 3 carboxylate ions in the form of their salts
- These salts are soap molecules

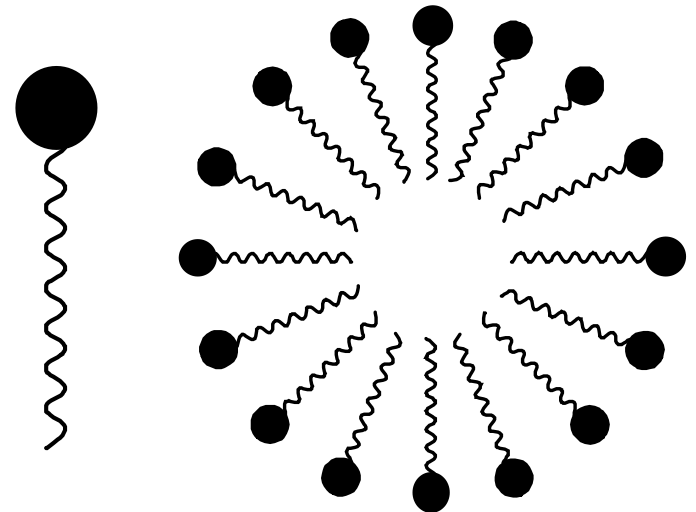
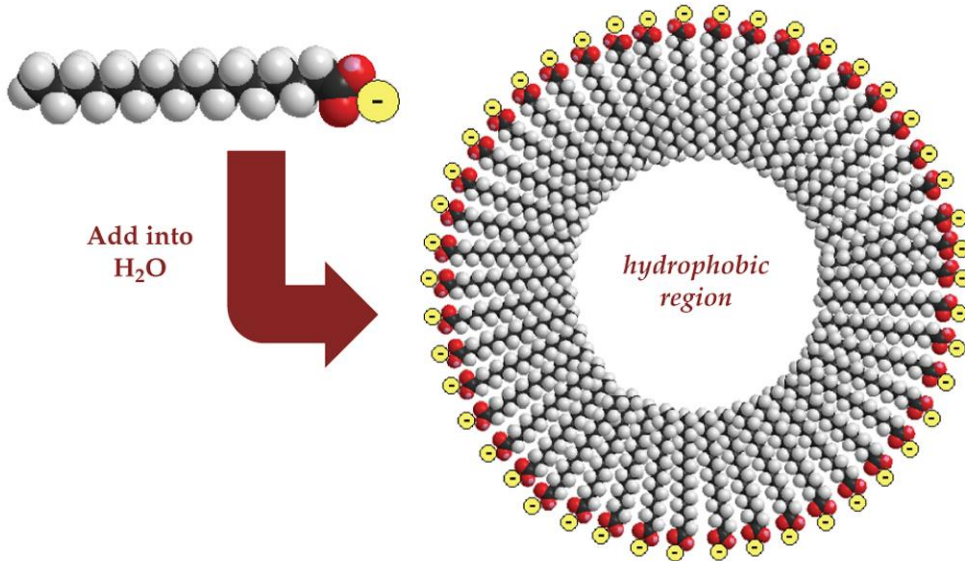


Soaps



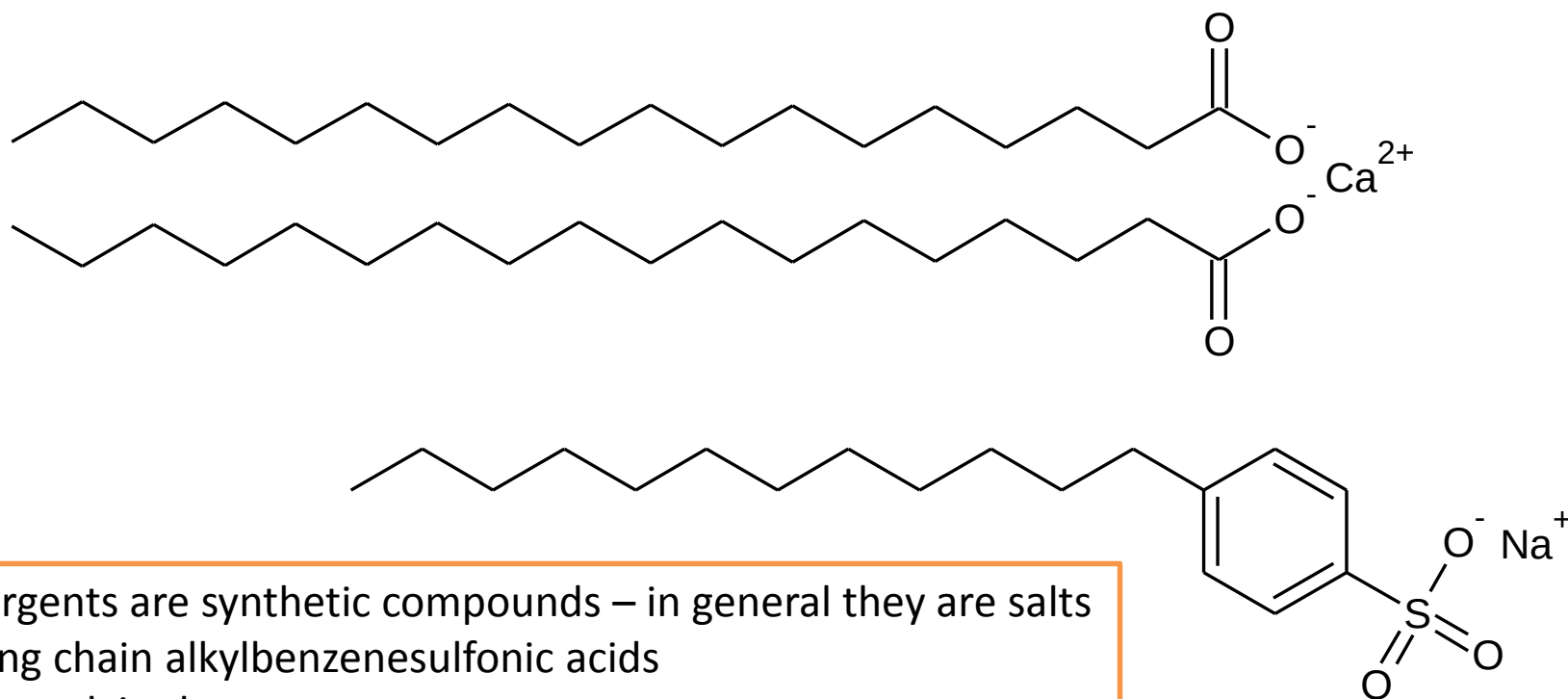
- A typical soap molecule is shown above
- Form micelles when added to water
- Grease and dirt aggregate in the hydrophobic region
- Oily substances become water soluble

The representations below are used to hand draw soap molecules and micelles



Detergents

- Hard water contains divalent cations such as Mg^{2+} and Ca^{2+}
- When soap is used in hard water 'soap scum' can form
- Soap scum is an insoluble Mg^{2+} or Ca^{2+} salt of a soap molecule



- Detergents are synthetic compounds – in general they are salts of long chain alkylbenzenesulfonic acids
- They work in the same way as soaps
- Their Mg^{2+} and Ca^{2+} salts are soluble, thus they do not form soap scum in hard water as soaps do